MUHAMMAD IMRAN Grand Quiz

30/ MARKS BT503 Environment Biotechnology

1 to 117 lessons for grand quiz (in lieu of Midterm examination)

107 MCQs

1. freshwater species *Chlamydomonas reinhardtii* was sensitive to _____ **all**
2. which of the following factors contribute to the increase in the death rate of the pathogenic microorganisms _____ **high DO concentrations and an increase in pH**
3. Plants are susceptible to bacterial pathogens such as *Erwinia carotovora*, which produces enzymes capable of degrading its _____ **cell walls**.
4. The average oceanic temperature is around _____ **1–3°C**
5. Which of the following bacteria causes crown gall disease _____ **Agrobacterium tumefaciens**
6. Which zone is essential in the treatment line for biological removal of phosphorus ____ **both of the above**
7. Salt tolerance in tomatoes has been established by introducing genes involved in …..antiport, the transport of sodium and hydrogen ions in opposite directions across a membrane_____ __ **hydrogen and sodium**
8. Those species which live in higher Kcl levels are known a_____ extremophiles _____ **halophile**
9. The process of expression of foreign genes in a plant is called _____ **Gene expression**
10. The removal of pollutants by plant that breaking down into compounds that can be vaporized is known as **----------. Degradation, phytovolatislation, phytosorption**
11. The removal of the organic matter in the secondary stage is carried out through biochemical reactions undertaken by. _____ **microorganisms**
12. Enzyme used in PCR is ….____-_____ **DNA Polymerase**
13. in facultative aerated lagoons are oxygen is supplied by mechanical equipment called aerators._____ _____ **aerators**
14. denaturation step in PCR usually occur at which temperature? _____ **95 ◦C**
15. Is thermophilic microorganism from with polymerase has been isolated and used for PCR purpose._____ **Thermus aquaticus**
16. many planktons at an increased rate in lakes when there are high centralization of __ **BOTH of the above .**
17. in bacteria the most likely candidates for genetic transfer are.._____ **both of the above**
18. the first working draft of the human genome sequence was published in which year ?_____ **2001**
19. which of the following is considered as health threatening microorganism of water _____ **Vancomycin-Resistant Enterococcus**
20. Halophiles are extremely important in environment biotechnology these are microorganisms that survive in highly saline environment . they do so by ensuring that**---**_____**-**

    -. Their cytoplasm contains a higher concentration of solutes then their surroundings
21. DNA is precipitated by _____ **Alcohol**
22. Genomic library are very useful to look for a specific Sequence of _____ **DNA**
23. Typically the soil types best suited to biotechnological interventions are _____**sands and gravels**
24. Which kind of porous medium is used in surface flow wetland_____
25. Which of the following chemicals is involved in the detoxification of the metals?
26. which of the following contaminants are rich in trichopria tetrat _____ **Phosphorus**
27. During nitrification PH level ?_____
28. Zooplanktons are microscopic ?
29. Wolffia globosa is an important tool for showing ….. sensitivity and contamination._____ **cadmium**
30. Multicellular organism cannot tolerate temperature….._____ **50 ◦ C**
31. In eukaryotes the first product of transcription form DNA is …

32. Bacillus thuringiensis tends to use codons richer in then the plant cells into which the gene is placed._____ **thymidine and adenine**
33. Bacteria have also adapted to degrade man-made organics called .
34. Polychlorinated biphenyls (PCBs) are xenobiotics having high level of ….
35. Biological treatment are not best suited to use in which type of soil .?
36. Optimum PH for local environment conditions lies in the range of ….?_____ **6.5–7.5**
37. Which of the following treatment system is adopted for removal of biodegradable organic matter form domestic sewage? _____ **aerobic biofilm reactors**
38. The sequence recognized by restriction enzyme is known as…_____ **recognition a site**
39. High rate of ponds usually receive a high ….. load per unit surface area?
40. Which of the following method may be employed to improve the efficiency of pollutant degradation by a bacterium without the use of genetic engineering?
41. Which of the following technique is most suitable for the detection of RNA
42. Which of the following agents should be removed from tertiary level of water treatment level?
43. Microbes and enzymes typically function at _____ **lower temperatures and pressures.**
44. Which of the following agent are used to support the formation of biofilm in low rate tricking filter?
45. Transgenic plants?
46. Which of the following chemical is removed from activated of sludge with biological nitrogen removal process?
47. The mechanism of separation of the particles with a density greater than sewage is adopted in …
48. The first bio tech based method of odor control was applied in..
49. Which gas is produced from up flow anaerobic sludge blanket reactor system?
50. The repeats flanking the T-DNA of agrobacterium tumefaciens are known as….
51. What is the main objective of maturation ponds?
52. In bacteria genetic transfer of material occur through …
53. Which of the following technique is most suitable for the detection of presence gene product.
54. A plasmid can be considered as a suitable cloning vector if?
55. which of the following is not characteristic of a transgenic crop?
56. What will be the sequences of application of waste water in wetlands in rainy season.
57. Main purpose of growing non food crop is to get …… commercially?
58. Which of the following group is now recognized as forming a distinct evolutionary line.
59. Restriction enzymes are enzymes
60. Which kind of filter is used in septic tank?
61. applications of the system overland flow is applicable to the …_____ **plain area**
62. bacillus thuringiensis tends to used codons richer in the plant cell into which the gene is placed.
63. generally the local environmental conditions required would ideally offer a temperature of.--- **20–30 ◦ C**
64. Which of the following statement is false?
65. The existence of microbes capable of surviving in extreme environments has been known since the.. **1960s**
66. *mer B* codes for which enzyme _____ **lyase**
67. a bacterial strain known to degrade PCBs is?_____ ***Pseudomonas putida***
68. lichens are used as bioindicators to monitor ……. Quality _____**air**
69. Extremozymes capable of functioning below pH1 have been isolated from these structures in some …species _____ **acidophile**
70. How many practical problems associated with GE of plants which make them more difficult to manipulate than bacteria _____ **2**
71. Of all soil which bacteria, seem to have the most highly developed ability to adapt quickly to new carbon sources _____ **Pseudomonads**
72. ….. are able to transfer genetic material between susceptible cells _____ **eukaryotic viruses**
73. Which this bacterium would later make possible the widespread use of a revolutionary technology, the polymerase chain reaction (PCR) _____ **Thermus aquaticus**

74. A bacterial gene encoding pentaerythritol tetranitrate reductase, an enzyme involved in the degradation of explosives, has been transferred into _____ **tobacco plant**

75. biological treatments are not best suited to in which type of soils _____ **all of the above**

76. The oestrogen is degraded completely to …….. thus presenting a plausible method for water polishing prior to consumption. _____ **carbon dioxide and water**

77. Of all soil bacteria …… seem to have the most highly developed ability to adapt quickly to new carbon sources. _____ **Pseudomonads**

78. Ralstonia eutropha, containing ……in its backbone, _____ **Sulphur**

79. A use of thermophilic enzyme is seen in the application pfu polymerase for PCR . pfu stands for _____ **Pyrococcus furiosus**

80. Reporter genes can only be a guide to the process of **transcription and translation** occurring in the cell _____-_____ **both b&c**

81. Hyperthermophiles work at temperatures above _____**85c**

82. Many surface structural proteins in halophiles require severely elevated concentrations of _____.

83. Isolated samples of common place proteins, like egg albumin, are irreversibly denatured well at which temperature? _____**blew 100c**

84. Bryophytes are frequently used to monitor_____**. air contamination**

85. The retention time of the solids in the system is called? _____ **sludge age**

86. Planktons are often used as indicators of water pollution. Many planktons reproduce at an increased rate in lakes when there are high centralizations of ____ and ___. _____ **phosphorus and nitrogen.**

87. When the bacteria starts to use organic material from the cellular material of extended aeration system, which kind of gas is produced ?

88. The removal of the organic matter in the secondary stage is carried out through biochemical reactions, undertaken by.

89. Imhoff tanks, are basically____tanks? _____ **sedimentation**

90. Plankton also plays an important role in biological deterioration _____ **organic matter**

91. In bacteria, the most likely candidates for genetic transfer are_____. Bacteriophage

92. Bacteria have also adapted to degrade man-made organics called _____**xenobiotics**.

93. Which of the following is not a characteristic of a transgenic crop?

94. What is the level of pH at which precipitation is done for removal of pollutants like phosphorus ?

95. Gene is expressed, preceded by a strong promoter, most commonly the '35S' promoter, obtained from_____. **Agrobacterium tumefaciens**

96. What is the main application of Biomemetics?

97. Which kind of plant genera are used in sub-suface flow system?

98. High rate ponds are conceived in order to maximize_____production?

99. Species found in the deep ocean volcanic vents known as 'black smokers' can tolerate_____?

100. T-DNA on Ti plasmid is flanked by a direct repeat of_____. **24bp**

101. What is the efficiency of maturation pounds for the removal of pathogenic organisms?_____ **99.99%**

102. Which statement about endocrine disruptors is TRUE?

103. In sub surface flow system rhizomes and roots are adjacent to which condition? _____ **Aerobic**

104. Glyphosate' (herbicide) is an analogue of phosphoenol pyruvate and inhibits the enzyme 5-enolpyruvylshikimate-3-phosphate synthase. The gene coding for this enzyme has been isolated and inserted into plants which then exhibited an increased resistant to the herbicide. The plants that gave the best results were: _____ **Petunias**

105. Microorganisms are often used as health indicators of aquatic and terrestrial ecosystems. e.g., Some microorganisms when exposed to cadmium and benzene contaminants produce _____ proteins, which can be used as early warning signs _____**stress**

106. To design production systems which avoid the potential for environmental contamination is the challenge of _____. **Green chemistry**

107. Pyrolobus fumarii, optimum temperature for reproduction is_____? **105 c**